

Problem Statement

Financial markets are highly dynamic, and predicting stock price movements is a challenging task influenced by numerous factors such as market trends, investor behavior, and global events. Investors and analysts often need a simple yet effective method to understand how a stock might behave in the near future based on its historical performance.

This project aims to develop a **Stock Trend Prediction Model** that uses **Linear Regression** to forecast the next 30 days of stock closing prices. By analyzing historical price data, calculating **average daily returns**, and assessing **market volatility**, the model provides insights into future trends that can assist in decision-making and market analysis.

Objectives

- To clean and preprocess stock data for accurate time-series analysis.
- To calculate average daily returns and volatility to understand price behavior.
- To apply **Linear Regression** for forecasting future stock prices.
- To visualize both historical and predicted trends for better interpretation.